Ian Chow

Astronomy Ph.D Student | University of Washington

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Education

PRESENT – Sep. 2025 | **University of Washington** — *Seattle, WA, USA*
Ph.D Astronomy
Supervisor : Prof. Mario Jurić

Aug. 2025 – Sep. 2023 | **University of Western Ontario** — *London, ON, Canada*
MSc. Astronomy
Thesis : Orbital and Physical Properties of Decameter-Sized Earth Impactors
Supervisor : Prof. Peter G. Brown

May 2023 – Sep. 2018 | **University of Toronto** — *Toronto, ON, Canada*
Honours BSc. w/ High Distinction – Astronomy & Physics Specialist, Statistics Major, Mathematics Minor
Astronomy Thesis : Analyzing Radial Velocity Data from the Resonant Planetary System HD 45364
Supervisors : Dr. Sam Hadden, Prof. Hanno Rein
Statistics Thesis : Probabilistic Dimensionality Reduction Methods for Stellar Chemodynamics
Supervisor : Prof. Joshua S. Speagle

Additional Research Positions

Aug. 2023 – May 2023 | **Dunlap Institute for Astronomy & Astrophysics, University of Toronto** — *Toronto, ON, Canada*
Summer Undergraduate Research Assistant
Project : Understanding the impact of Bayesian inference on ultra-light axion limits
Supervisors : Dr. Keir K. Rogers, Prof. Renée Hložek

Aug. 2022 – May 2022 | **Canadian Institute for Theoretical Astrophysics (CITA)** — *Toronto, ON, Canada*
Summer Undergraduate Research Fellow
Project : Modelling Migration Scenarios of Resonant Planets Using Radial Velocity Data
Supervisors : Dr. Sam Hadden, Prof. Hanno Rein, Prof. Norman Murray

Peer-Reviewed Journal Publications

First Author

1. **Chow, I.**, Jurić, M., Jones, R.L., Kiker, K., Moeyens, J., Brown, P.G., Heinze, A.N., & Kurlander, J.A. "Predictions of Imminent Earth Impactors Discovered by LSST." 2026, *ApJ*, 1001(1), 61.
   Media Coverage/Press Release : Universe Today (**YouTube Interview**, **Article**), **Sky and Telescope**, **University of Washington**
2. **Chow, I.**, & Brown, P.G. "Decameter-sized Earth Impactors – II : A Bayesian Inference Approach to Meteoroid Ablation Modeling." 2026, *JGR : Planets*, 131, e2025JE009392.
3. **Chow, I.**, & Brown, P.G. "Decameter-sized Earth impactors – I : Orbital properties." 2025, *Icarus*, 429, 116444.
4. **Chow, I.**, & Hadden, S. "Influence of Modeling Assumptions on the Inferred Dynamical State of Resonant Systems : A Case Study of the HD 45364 System." 2025, *ApJ*, 980(2), 236.

Second/Third Author

1. Chandler, C.O., Murtagh, J., **Chow, I.**, West, M., Gillan, A.F., Ferguson, P.S., Kelley, M.S.P., Kurlander, J.A., Hsieh, H.H., Bernardinelli, P.H., Beebe, W., Kubica, J., Connolly, A., & Davenport, J.R.A. "Tails and Trails : Comets and NEOs Survive Co-addition in Rubin Early Data Preview 2." In review at *ApJL*.

Contributing Author

1. Cheng, Q., Scolnic, D., Kurlander, J.A., **Chow, I.**, & Fernandes, M.B. "Assessing the Vera Rubin Observatory's Ability to Discover Asteroid Impactors Before They Collide with Earth." 2026, *AJ*, 171(5), 309.

Research Notes & White Papers

Contributing Author

1. Hsieh, H.H. et al. (25 co-authors including **Chow, I.**) "The 2026 Reactivation of Main-belt Comet 433P/(248370) 2005 QN_{173}." 2026, *RNAAS*, 10(8), 228.
2. Singh, D. et al. (30 co-authors including **Chow, I.**) "A Tail of Two Classifications : Citizen Science Reveals Activity from Jupiter-family Comet 2021 RK_{133}." 2026, *RNAAS*, 10(7), 210.

Selected Awards, Scholarships, Fellowships & Honours

2026	**Astronomy Department Graduate Student Research Prize**, $1,000 USD	*University of Washington*
2026	**Schweickart Prize Honorable Mention**, honour	*B612 Foundation*
2025	**Graduate School Top Scholar Award**, $10,830 USD	*University of Washington*
2025-2028	**NSERC Postgraduate Scholarship – Doctoral (PGS-D)**, $120,000 CAD	*NSERC*
2024	**NASA International Space Apps Challenge Global Winner**, honour	*NASA*
2024-2025	**Ontario Graduate Scholarship (OGS)**, $15,000 CAD	*University of Western Ontario*
2023-2025	**Western Graduate Research Scholarship**, $13,137 CAD	*University of Western Ontario*
2023	**SURP Symposium Poster Competition Award**, $50 CAD	*University of Toronto*
2023	**Summer Undergraduate Research Program (SURP) Award**, $9,980 CAD	*University of Toronto*
2022	**Summer Undergraduate Research Fellowship (SURF)**, $9,500 CAD	*CITA*
2022	**Smith Solis Research Scholarship in Astronomy and Astrophysics**, $1,250 CAD	*University of Toronto*
2020-2023	**Dean's List Scholar**, honour	*University of Toronto*

Observing Programs

Observing Time Awarded Summary

2026D	**Dominion Astrophysical Observatory Plaskett 1.8 m**, 5 nights	*PI*
	Photometric Characterization of Small Near-Earth Objects in the LSST Era	
2026-	**Apache Point Observatory ARC 3.5 m**, 9 half-nights (PI) + 42 half-nights (Co-I)	*PI + Co-I*
	Discovery and follow-up characterization of small Solar System bodies	
2026B	**Gemini Observatories Gemini South 8.1 m**, 9.8 hours	*Co-I*
	Near-Infrared Spectroscopic Characterization of Sub-140 m Near-Earth Asteroids with Gemini/FLAMINGOS-2 : A Pilot Survey for the LSST Era	

Observing Experience

2025-2026	**Apache Point Observatory** ARC 3.5 m (IAU code 705), 31 half-nights, remote	*Sunspot, NM, USA*
2026	**Višnjan Observatory** Dagor 1.0 m (IAU code L01), 10 nights, on-site	*Tićan, Croatia*

Selected Academic Presentations

Contributed Conference Talks

Oct. 2026	**58th Annual DPS Meeting**, AAS Division for Planetary Sciences	*Spokane, WA, USA*
Aug. 2026	**Monsoon Workshop 2026**, Catalina Sky Survey	*Tucson, AZ, USA*
Jul. 2026	**Asteroids Comets Meteors 2026**, Adam Mickiewicz University	*Poznań, Poland*
Sep. 2025	**EPSC-DPS 2025**, Europlanet/AAS Division for Planetary Sciences	*Helsinki, Finland*
Jul. 2025	**Meteoroids 2025**, Curtin University	*Perth, WA, Australia*
Mar. 2025	**56th Lunar and Planetary Science Conference**, NASA/Lunar and Planetary Institute	*The Woodlands, TX, USA*
May 2024	**55th Annual DDA Meeting**, AAS Division on Dynamical Astronomy	*Toronto, ON, Canada*
Aug. 2022	**2022 CITA Planet Day**, Canadian Institute for Theoretical Astrophysics	*Toronto, ON, Canada*

Contributed Conference Posters

Jun. 2024	**2024 CASCA Annual General Meeting**, University of Toronto/York University	*Toronto, ON, Canada*
Jun. 2023	**2023 CASCA Annual General Meeting**, NRC Herzberg/University of British Columbia	*Penticton, BC, Canada*

Other Invited Presentations

Apr. 2026	**Center for Astrophysics and Cosmology Graduate Seminar**, University of Nova Gorica	*Ajdovščina, Slovenia*
Feb. 2026	**Cosmology Group Seminar**, Duke University	*Durham, NC, USA*
Jun. 2025	**NASA Funding Review**, NASA Meteoroid Environment Office (remote)	*Huntsville, AL, USA*
Feb. 2025	**CSA Executive Committee Meeting**, Canadian Space Agency (remote)	*Virtual*
May 2024	**NASA Funding Review**, NASA Meteoroid Environment Office (remote)	*Huntsville, AL, USA*

Public Outreach Talks

Mar. 2026	**"Great Balls of Fire : Meteors and Meteorites in the Past, Present and Future"**, Astronomy on Tap Seattle	*Seattle, WA, USA*
Aug. 2025	**"Brighter than the Sun : Fireballs in Earth's Atmosphere"**, Hume Cronyn Memorial Observatory	*London, ON, Canada*
Apr. 2025	**"SkyShield : Protecting Earth and Space Infrastructure From Space Hazards"**, Royal Astronomical Society of Canada London	*London, ON, Canada*

Teaching Experience

I instructed students on how to operate and acquire, reduce, and process data from the Višnjan Observatory's 1.0 m Dagor telescope on-site in Tićan, Croatia, helping confirm 40+ new near-Earth objects published in the *Minor Planet Electronic Circulars*.

2026 **UW-Višnjan School of Astronomy**, Teaching Assistant — *U. of Washington/Višnjan Obs.*

My duties in the following course included delivering lectures, conducting in-class demonstrations, holding office hours, and proctoring, grading and reviewing exams.

2024-2025 **Astronomy 1021 : General Astronomy**, Teaching Assistant (x2) & Guest Lecturer — *University of Western Ontario*

My duties in the following course included supervising lab sessions and grading lab reports.

2023-2024 **First-Year Physics Labs**, Teaching Assistant (x2) — *University of Western Ontario*

My duties in the following courses included delivering in-person tutorials and help centres, running midterm viewing sessions, and proctoring, grading and reviewing exams.

2024 **Physics 1402 : Physics for Engineering Students II**, Teaching Assistant — *University of Western Ontario*
2023 **Physics 1201 : Physics for the Sciences I**, Teaching Assistant — *University of Western Ontario*

Advising Experience

I have advised the following students :

2026- **Sanja Dumenčić**, UNG INSPIRO 2.0 project (co-advised with Mario Jurić) — *University of Nova Gorica*
Early Detection of Potentially Hazardous Asteroids with the Vera C. Rubin Observatory

Other Professional Experience

Sep. 2020 – Jun. 2020 **Innovere Medical** — *Markham, ON, Canada*
Software Developer

- Automated detection of dropouts in time-series audio data from an MRI scanner's wireless audio system using power spectrum analysis in MATLAB and Python, eliminating 20+ hours of work weekly
- Developed and tested TechSmart, an in-house multimedia app for patient use during MRI scans, with company's software development team

MATLAB | Python | Signal Processing

Aug. 2019 – Jun. 2019 | **Plantiga Technologies** — *Vancouver, BC, Canada*
Software Developer

- Developed methods to compute physical fitness heuristics from time-series acceleration (g-force) data, using signal processing techniques like digital filtering and convolution in Python (NumPy, SciPy, Pandas) to improve detection of foot impacts
- Field-tested and validated hardware such as sensor shoe insoles that track movement
- Acquired data from company partners such as physiotherapy clinics, universities (University of British Columbia, Simon Fraser University), and sports organizations (Houston Rockets, US Tennis Association)
- Wrote documentation of company products and services for clients

Python | Signal Processing | Data Analysis

Aug. 2017 – Jun. 2017 | **Synced Review** — *Toronto, ON, Canada*
Research Intern

- Conducted literature review focusing on advancements in reinforcement learning used in adversarial-search board and video game artificial intelligence programs for a company report
- Worked with company team to research and edit review articles on industry trends in machine learning and robotics technology

Machine Learning | Artificial Intelligence | Literature Review

Selected Leadership, Volunteering & Service

PRESENT – Feb. 2026 | **Apache Point Observatory** — *Seattle, WA, USA*
Telescope Allocation Committee (TAC) Graduate Student Representative

- Served as the University of Washington's graduate student representative for allocating time at the Apache Point Observatory ARC 3.5-meter telescope
- Read and evaluated observing proposals to equitably allocate 40+ nights quarterly in collaboration with faculty and postdoctoral members of the TAC

Aug. 2025 – Jun. 2024 | **Hume Cronyn Memorial Observatory** — *London, ON, Canada*
Outreach Volunteer

- Volunteered at astronomy Public Nights attended by 80+ visitors weekly at the University of Western Ontario's Cronyn Observatory

May 2023 – Jan. 2019 | **University of Toronto Academic Trivia Club** — *Toronto, ON, Canada*
Vice President, Competitor, Tournament Organizer & Question Writer/Editor

- Elected Vice President of the University of Toronto's Academic Trivia Club during the 2020-2021 and 2021-2022 academic years organizing twice-weekly practices and social events, managing club social media presence, and moderating practices and tournament games
- Organized and directed several collegiate and high school tournaments, including the 2021 University of Toronto Collegiate Novice and the 2022 University of Ottawa ACF Fall tournaments, played by 30+ collegiate teams in total across Canada and the U.S.
- Wrote and edited trivia questions across a wide range of academic disciplines (including astronomy and physics) for 2022 WORKSHOP, 2023 Canadian Novice, and 2024 MRNA III, collegiate tournaments played by 80+ teams in total across Canada, the U.S., and the U.K.

Software Projects

SkyShield Orrery 2024

spaceapp-wmpgang2024orrery.netlify.app/

An interactive, physics-based digital Solar System orrery highlighting near-Earth objects and meteoroids. Developed by myself, Dakota Cecil, Simon Van Schuylenbergh and Maximilian Vovk for the 2024 NASA International Space Apps Challenge and selected by NASA as one of 10 Global Winners out of 10,000 submitted projects.

Press Release : University of Western Ontario

HTML | CSS | JavaScript

fasano-franceschini-test 2024

github.com/wmpg/fasano-franceschini-test

A pure Python implementation for the multivariate two-sample Kolmogorov-Smirnov (K-S) statistical test described by Fasano & Franceschini (1987) and Puritz et al. (2023). Published as part of Chow & Brown (2025).

Python

Skills

Programming	Python (NumPy, SciPy, Astropy, Pandas, Matplotlib, Keras/TensorFlow, scikit-learn), MATLAB, R (ggplot, dplyr), HTML5 (Bootstrap), CSS, JavaScript (Node.js)
Software	LaTeX, Git/GitHub, Jupyter Notebook, Anaconda, R Suite, Bash, Linux, SAOImageDS9, Microsoft Excel

Other Affiliations and Organizational Memberships

2025-PRESENT	LSST Solar System Science Collaboration
2025-PRESENT	American Astronomical Society
2025-PRESENT	Europlanet Society
2023-PRESENT	Canadian Astronomical Society